RESEARCH PAPER PRESENTATION

Benchmarks Are Not That Out of Distribution: Word Overlap Predicts Performance

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February 2026

Language Models | Benchmark Analysis | Data Quality

OVERVIEW

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CHAPTER 01

Introduction & Research Background

01

Understanding the relationship between pre-training data and benchmark performance

INTRODUCTION

The Central Question in LM Training

The Challenge

Understanding what constitutes high-quality pre-training data remains a central and unresolved question in language model training. While improving benchmark scores is a primary objective, which datasets best translate to strong downstream results is not yet fully understood.

Benchmark Assumptions

Benchmarks are constructed around representative problem formulations for systematic quantitative assessment. The framework **implicitly assumes** that improved benchmark scores reflect a more profound understanding of task-relevant knowledge.

Empirical Correlation

Models achieving **low training loss on high-quality datasets consistently attain superior benchmark performance** . This correlation can be interpreted as implying that strong downstream performance arises from effective learning of statistical patterns in training data.

Key Insight: Compression-Based Generalization

Compression View
Language modeling as compression

Pattern Learning
Learning statistical patterns in data

Distribution Overlap
Critical factor for performance

HYPOTHESIS

Research Hypothesis & Key Finding

Central Hypothesis

Benchmark performance is **primarily driven by the degree of statistical pattern overlap** between pre-training corpora and evaluation datasets.

Metric 1

Word-Level Unigram Cross-Entropy

Measures how well the word frequency distribution of the pre-training corpus predicts that of the benchmark dataset, capturing their marginal distributional similarity.

Metric 2

Word Frequency Statistics

Quantifies the amount of statistical support underlying the distributional alignment, complementing cross-entropy by reflecting absolute word frequencies.

Key Finding

There is a **robust inverse relationship** between word-level unigram cross-entropy and benchmark performance. Lower cross-entropy indicates closer alignment between training and benchmark word distributions, implying substantial statistical overlap. This suggests that **many standard benchmarks are only weakly out-of-distribution** relative to pre-training corpora.

CONTRIBUTIONS

Contributions of This Work

1

Demonstrating Word Overlap Alignment

We demonstrate that **benchmark scores are strongly aligned with how closely the word overlap** of a pre-training corpus matches that of the benchmark. This provides empirical evidence for the relationship between distributional similarity and downstream performance.

2

Introducing Novel Proxy Metrics

We introduce **word-level unigram cross-entropy and word-frequency statistics** as simple, tokenizer-agnostic proxy metrics for quantifying dataset overlap. These metrics avoid biases introduced by subword tokenization and provide clean measures of marginal distribution overlap.

3

Controlled Experimental Validation

Through **controlled comparisons across 10 benchmarks, 4 pre-training datasets, and 3 model scales** , we show these proxies reliably predict how pre-training data choices translate into benchmark scores.

Benchmarks	Datasets	3 Model Scales

CHAPTER 02

Motivation & Theoretical Framework

02

Information-theoretic perspective on benchmark performance

THEORETICAL FRAMEWORK

The Information Theory Perspective

Out-of-Distribution Quantification

KL Divergence Framework

Out-of-distribution (OOD) can be quantified by the divergence **$D_{KL}(Q||P)$** where:

- **P** = training data distribution
- **Q** = benchmark distribution

Small Divergence Implication

When strong benchmark performance accompanies low training loss, this implies a **small divergence** , indicating that the benchmark distribution is statistically close to the training data and thus **only weakly OOD** .

Language Modeling as Compression

Minimizing next-token prediction loss is **equivalent to minimizing the codelength** required to encode the data under an optimal entropy code. Language models operate analogously to **universal compressors** that encode regularities in the training data.

Key Insight: A strong correlation between training loss and downstream performance suggests that the benchmark and training data are **nearly in-distribution** in an information-theoretic sense, since a statistical compressor can operate with similar effectiveness only when the two datasets share substantial overlap in their underlying statistical patterns.

The Core Research Question

Is the degree of word overlap between training and benchmark data heavily influencing benchmark performance trends?

COMPRESSION VIEW

Compression & Generalization

The Compression-Based Generalization View

Training Data
Source distribution P

Compression
Learn statistical patterns

Benchmark
Target distribution Q

Compression-based generalization transfers most reliably when the benchmark distribution is **close to the source distribution (in-distribution)** . A statistical compressor can operate with similar effectiveness (e.g., exhibiting similar compression ratios) only when the two datasets share substantial overlap in their underlying statistical patterns.

When Distributions Align

- Low training loss on P
- Strong benchmark performance on Q
- Small KL divergence $D_{KL}(Q||P)$
- **Weakly out-of-distribution**

The Critical Question

Clarifying this mechanism is essential for understanding the characteristics of **high-quality pre-training datasets** that yield the best language models. More broadly, it invites us to rethink what constitutes a truly diagnostic benchmark — one that measures generalization beyond superficial distributional affinity with the training corpus.

Implication: If benchmark performance is primarily driven by word overlap, then **simply learning statistical patterns present in the pre-training corpus can substantially contribute to improved benchmark performance** , challenging the interpretation of benchmark gains as evidence of deep understanding or generalization.

CHAPTER 03

Methodology

Measuring statistical pattern overlap between datasets

03

PROXY METRICS

Proxy Metrics for Dataset Overlap

Two Complementary Metrics

1 Word-Level Unigram Cross-Entropy

Measures how well the word frequency distribution of the pre-training corpus predicts that of the benchmark dataset, thereby capturing their **marginal distributional similarity**.

$$H(P_{eval}, P_{pre}) = -\Sigma\, P_{eval}(w) \log P_{pre}(w)$$

2 Word Frequency Statistics

Quantifies the **amount of statistical support** underlying the distributional alignment captured by unigram cross-entropy. This is important because cross-entropy depends only on normalized marginals and does not reflect absolute word frequencies.

Computation Process

Tokenization
Simple whitespace splitting

No Normalization
No lowercasing or NFKC

Frequency Est.
Empirical word frequencies

Smoothing
Laplace (add-one) smoothing

Why Laplace Smoothing? Words that are absent from the pre-training corpus but appear in the benchmark dataset need to be handled. Laplace (add-one) smoothing ensures every word has non-zero probability, preventing infinite cross-entropy values.

RATIONALE

Why Word-Level Unigram Cross-Entropy?

Reason 1: Avoiding Tokenization Bias

Tokenization bias arises because subword tokenizers are many-to-one mappings that transform the underlying byte-level probability space. Although a tokenized language model is statistically equivalent to its byte-level counterpart at the sequence level, its conditional distributions differ due to constraints imposed by token boundaries.

Conditioning on tokenized text **discards probability mass** of alternative valid encodings under the byte-level conditional distribution.

The Problem

Token-level cross-entropy **conflates genuine distributional mismatch** with segmentation artifacts introduced by the tokenizer.

The Solution

Word-level unigram cross-entropy operates at a granularity that **more faithfully reflects empirical frequency alignment** between datasets.

Reason 2: Higher-Order N-gram Issues

For $n \geq 2$, an n-gram model restricts the conditional distribution to an order-(n-1) Markov form, thereby **projecting the true data-generating process onto a misspecified hypothesis class**.

$$H_n(B|A) = H(P_B) + D_{KL}(P_B \,||\, P_{A(n)})$$

The KL term splits into two components: **Markov misspecification** (information lost when long-range dependencies are truncated) and **dataset mismatch** (differences between truncated conditionals).

The misspecification term persists even when A = B and therefore dominates the cross-entropy for realistic n, masking differences due to genuine distributional mismatch. This is further exacerbated by exponential growth in sample complexity.

CHAPTER 04

Experimental Settings

04

Comprehensive evaluation across datasets, models, and benchmarks

TRAINING

Training Configuration

Pre-training Datasets

- **FineWeb-Edu** — Recent high-quality corpus
- **DCLM** — Recent high-quality corpus
- **C4** — Earlier high-quality corpus
- **OpenWebText** — Lower-quality corpus

Model Architectures

All models use **LLaMA architecture** with GPT-2 tokenizer. Three model scales evaluated:

400M Parameters	24 layers, 8 heads
1.33B Parameters	30 layers, 30 heads
3.36B Parameters	32 layers, 20 heads

Training Hyperparameters

Optimizer	β_1, β_2	Weight Decay	Grad Clip	LR Schedule	Warmup
AdamW	0.9, 0.95	0.1	1.0	Cosine	350M tokens

8.5B	26B	60B
Minimum Tokens	Medium Tokens	Maximum Tokens

EVALUATION

Evaluation Benchmarks

We evaluate on **10 representative benchmarks**, all in zero-shot setting, covering diverse reasoning capabilities:

ARC Easy Science QA (Easy)	**ARC Challenge** Science QA (Hard)	**Hellaswag** Commonsense	**MMLU** Multitask	**SciQ** Science QA
OpenBookQA Open-book QA	**PIQA** Physical Reasoning	**LAMBADA** Long Context	**SocialIQA** Social Reasoning	**SWAG** Situations

Evaluation Metrics

- **Accuracy** for most benchmarks (higher is better)
- **Perplexity** for LAMBADA (lower is better)
- All evaluations in **zero-shot setting**

Benchmark Coverage

- Knowledge-heavy tasks (ARC, MMLU, SciQ)
- Commonsense reasoning (Hellaswag, PIQA, SWAG)
- Reading comprehension (OpenBookQA, LAMBADA)
- Social reasoning (SocialIQA)

Comprehensive Coverage: These benchmarks span diverse reasoning capabilities including scientific knowledge, commonsense reasoning, reading comprehension, and social intelligence, providing a robust evaluation of language model performance across different domains.

CHAPTER 05

Main Results

Empirical evidence for word overlap predicting performance

05

MAIN FINDING

Word Overlap Explains Performance Trends

Core Finding

We find a **consistent negative correlation** between benchmark performance and unigram cross-entropy across all 10 benchmarks. Performance improves as word overlap between pre-training corpora and evaluation benchmarks increases, particularly when words in the benchmark dataset occur more frequently in the pre-training data.

The Stable Pattern

For every benchmark, the **ordering of pre-training corpora by word-level cross-entropy exactly mirrors the ordering by downstream score.** The dataset whose unigram distribution is closest to the benchmark (lowest cross-entropy) consistently produces the best performance.

Example from Figure 1 (3.36B model, 60B tokens):

- **ARC Easy:** FineWeb-Edu (11.19) > DCLM (11.55) > C4 (11.65)
- **MMLU:** FineWeb-Edu (11.36) > DCLM (11.41) > C4 (11.52)
- **Hellaswag:** C4 (11.04) > DCLM (11.06) > FineWeb-Edu (11.20)

Robustness Across Settings

This trend holds across diverse experimental configurations:

Dataset Sizes
8.5B, 26B, and 60B tokens

Model Scales
400M, 1.33B, and 3.36B parameters

Benchmarks
10 diverse evaluation tasks

Key Insight: This relationship holds both in settings where high-quality corpora dominate (e.g., ARC, MMLU) and where they do not (e.g., Hellaswag, PIQA where C4 performs best). **Word-level cross-entropy serves as a reliable, task-dependent proxy** for quantifying dataset overlap between pre-training corpora and evaluation benchmarks.

MECHANISTIC ANALYSIS

Coverage Distribution Analysis

Decomposing Word Overlap

Table 1 provides a mechanistic explanation by decomposing overlap into **coverage distributions** across different frequency regimes. This reveals that score differences are driven not by word coverage but by **how probability mass is allocated** across coverage regimes.

~80%	~95%	95-100%	Seen
High-frequency words	Medium-frequency words	Long-tail words	Total coverage

Nearly Identical Coverage

All pre-training datasets achieve **nearly identical seen word coverage (~99.9%)** for ARC Easy, MMLU, and Hellaswag. This **rules out word coverage** as a meaningful explanatory factor for performance differences.

Probability Mass Allocation

Score differences closely track entropy. FineWeb-Edu generally achieves the **lowest cross-entropy and highest scores** , while C4 exhibits higher entropy and weaker performance. High-performing datasets concentrate more mass in **high-frequency regions** and less in the long tail.

Example: ARC Easy (1.33B model, 26B tokens)

FineWeb-Edu	DCLM	C4
Entropy: 11.19	Entropy: 11.55	Entropy: 11.65
Score: 65.87	Score: 63.17	Score: 54.55

WORD FREQUENCY

Impact of Word Frequency Statistics

Beyond Normalized Distributions

Unigram cross-entropy is computed from **normalized word frequencies** and therefore abstracts away the absolute scale of dataset overlap. Word frequency statistics quantify the **amount of statistical support** underlying the distributional alignment captured by unigram cross-entropy.

Why This Matters: Language model pre-training operates under a **highly class-imbalanced regime** where word frequency directly determines the strength of the learning signal. Frequent words receive substantially more parameter updates, allowing their representations to converge rapidly to low loss, whereas rare words remain poorly estimated.

Table 2: Scaling from 8.5B to 26B Tokens

Training 400M models on different subset sizes reveals:

Cross-Entropy: Invariant
Remains constant within each dataset when scaling data

Performance: Improves
Consistently improves with larger token counts

Interpretation

Greater token exposure **strengthens the learning signal** for words in benchmark datasets, enabling more reliable convergence for frequent terms and better coverage of the long tail. This indicates that word frequency statistics capture an essential aspect of data overlap that **complements unigram cross-entropy** by reflecting the effective learning signal available during pre-training.

Example: ARC Easy Performance Gains

FineWeb-Edu
8.5B: 57.00% → 26B: 62.42%

DCLM
8.5B: 51.56% → 26B: 56.78%

C4
8.5B: 45.16% → 26B: 50.38%
+5.22 points

Cross-entropy remains at 11.19, 11.55, 11.65 respectively for both sizes

ROBUSTNESS

Consistent Trends Across Training Subsets

Testing Subset Independence

To examine whether the negative correlation depends on specific subset pairings, we compute cross-entropy across **five independently sampled 8.5B token subsets** from each pre-training dataset and pre-train 400M models on each subset.

Figure 2: Minimal Subset Variation

When pre-training data scale is controlled, model performance shows **no meaningful dependence** on the choice of data subset. Benchmark scores vary only marginally across subsets, with standard deviations small relative to absolute performance differences between datasets.

Table 3: Highly Consistent Metrics

Coverage distributions and word-level cross-entropy remain **highly consistent across subsets** within each dataset, indicating that marginal word frequency distributions are preserved under subset sampling when data scale is fixed.

Key Finding: Stability Across Subsets

These fluctuations are **consistent with run-to-run noise** and do not indicate systematic sensitivity to subset composition. Once data scale is controlled, benchmark performance is largely insensitive to subset sampling, and observed differences between pre-training datasets reflect **stable distributional properties** rather than stochastic effects arising from particular subset choices.

ARC Easy (Mean ± Std)

FineWeb-Edu	56.35 ± 0.368
DCLM	50.76 ± 0.505
C4	46.44 ± 0.713

PIQA (Mean ± Std)

FineWeb-Edu	67.48 ± 0.420
DCLM	66.35 ± 0.120
C4	65.59 ± 0.637

CHAPTER 06

Further Analysis

06

Benchmarks where word overlap does not explain performance

EXCEPTIONS

Grammar Benchmarks: BLiMP

BLiMP: The Benchmark of Linguistic Minimal Pairs

BLiMP evaluates **linguistic acceptability** through minimal pairs designed to probe grammatical phenomena. The benchmark consists of **67 sub-datasets**, each containing 1,000 minimal pairs that isolate contrasts in syntax, morphology, or semantics.

⚠ Complete Coverage

Owing to its **controlled vocabulary**, every word appearing in the benchmark is observed across all pre-training corpora. This results in **100% word coverage**, ruling out word coverage as a confounding factor.

Higher-Order Knowledge

Variations in word-level frequency statistics alone are **insufficient to explain performance differences** on BLiMP. Success reflects the acquisition of **grammatical knowledge grounded in higher-order, contextual representations** rather than surface-level word overlap.

Table 4: BLiMP Results (1.33B model, 26B tokens)

DCLM	C4	FineWeb-Edu
Entropy: 12.42 Score: 61.01	Entropy: 12.66 Score: 80.48	Entropy: 12.87 Score: 80.89

Note: Higher entropy correlates with higher performance — opposite of the main trend!

Insight: BLiMP demonstrates that some tasks require understanding beyond surface-level statistical patterns. Grammatical knowledge must be acquired through **contextual learning and structural understanding** rather than mere word frequency alignment.

EXCEPTIONS

Math Benchmarks: MathQA

MathQA: Mathematical Word Problem Solving

MathQA is designed to evaluate a model's ability to perform **arithmetic reasoning** and solve mathematical word problems that require **multi-step numerical computation.**

Unbounded Numerical Space

Unlike natural language benchmarks, mathematical expressions draw from an **effectively unbounded numerical space,** making direct coverage through pre-training inherently limited.

Abstract Rule Understanding

Success requires understanding **abstract arithmetic rules and operations** rather than relying on surface-level pattern compression.

Why MathQA Is Different

Compositional

Multi-step reasoning required

Algorithmic

Systematic procedures needed

Sparse Coverage

Rarely represented in pre-training

Performance Pattern

Zero-shot performance on math benchmarks is constrained by the need to **generalize beyond observed patterns to systematic reasoning procedures.** This helps explain why many mathematical benchmarks remain challenging in zero-shot settings: the underlying solution patterns are compositional, algorithmic, and sparsely represented in pre-training data.

Table 4 Results (1.33B model, 26B tokens): MathQA scores are consistently low (~22-23%) across all datasets with minimal variation, despite different cross-entropy values. This confirms that word overlap is not the primary driver of performance on mathematical reasoning tasks.

EXCEPTIONS

Multilingual Zero-Shot Evaluation

The Multilingual Challenge

Multilingual zero-shot evaluation refers to training primarily on **monolingual data** and evaluating on **unseen languages**. This scenario has attracted substantial attention as high-quality low-resource datasets remain scarce for large-scale multilingual training.

Hypothesis: Because modern pre-training corpora are rarely strictly monolingual even after language filtering, performance differences across unseen languages may arise from the **non-monolingual composition** of the training data and its implicit language mixture. If multilingual zero-shot gains were primarily driven by such effects, one would expect word-level cross-entropy to correlate with zero-shot performance.

Contrary to Expectations

Figure 3 shows that **word-level cross-entropy fails to capture any consistent trend** in multilingual performance. Neither the expected negative correlation for PIQA accuracy nor the expected positive correlation for LAMBADA perplexity is observed.

Evaluation Setup

- **Model:** 3.36B parameters
- **Training Data:** 60B DCLM tokens
- **Benchmarks:** Global PIQA, LAMBADA multilingual
- **Languages:** Arabic, Chinese, French, German, Italian, Portuguese, Spanish

Key Finding

Multilingual zero-shot performance cannot be straightforwardly explained by word-level distributional overlap alone. This suggests that cross-lingual transfer involves more complex mechanisms beyond simple word frequency alignment, possibly including structural similarities, shared subword patterns, or deeper semantic representations that transcend surface-level vocabulary overlap.

CHAPTER 07

Discussion & Implications

07

Broader implications for benchmark design and data quality

DISCUSSION

Possibility of Benchmark Performance Hacking

A Concerning Implication

The observed correlation between word-level overlap and benchmark performance suggests that **pre-training data can be modified in a controlled manner to improve downstream results** without necessarily improving true understanding or generalization.

Strategy 1: Synthetic Data Generation

Design a **synthetic dataset** where each sentence is generated to include words that are rare in the pre-training corpus but appear in the benchmark, using large language model APIs.

Approach:

- Identify rare benchmark words
- Generate sentences containing them
- Replace portion of pre-training data

Strategy 2: Vocabulary-Constrained Generation

Construct a synthetic dataset by **penalizing the logits of words outside the benchmark vocabulary** during text generation, then use for continual pre-training.

Approach:

- Constrain generation to benchmark vocab
- Generate diverse sentences
- Use for continual pre-training

A Principled Alternative

These approaches offer a **controllable and principled alternative** for constructing higher-quality pre-training subsets. However, they also raise important questions about the validity of benchmark evaluations and the need for more robust measures of true understanding and generalization.

Ethical Consideration: While these strategies could improve benchmark scores, they may not lead to genuinely better language models. The research community must develop **benchmarks that are resistant to such manipulation** and truly measure generalization capabilities.

DISCUSSION

Impact of Tokenizer on Performance

Tokenization and Representation Learning

Contextual representations are learned over **tokenized text** rather than raw text. Accordingly, different tokenizers can yield different performance even on the same dataset, and varying vocabulary size under a fixed corpus can substantially affect performance.

Tokenizer Matters

Different tokenizers → different performance

Vocabulary Size

Affects unigram loss and theoretical limits

Whitespace Pre-tokenization

Modern tokenizers approach word-level

Why Word-Level is Reasonable

- Modern tokenizers based on **whitespace pre-tokenization**
- With large vocabularies, approach **word-level tokenization**
- Same granularity as unigram cross-entropy measurement
- **BPE is lossless compression** preserving all information

Approximation Quality

Quantifying statistical pattern overlap at the word level provides a **reasonable approximation** for understanding downstream performance, which is largely driven by learned contextual representations. While not perfect, word-level analysis captures the dominant factors influencing benchmark scores.

Theoretical Connection: Vocabulary size under a fixed corpus affects performance through unigram loss and theoretical limits related to **Kolmogorov complexity**. However, in practice, the word-level approximation works well for understanding the relationship between data overlap and benchmark performance.

IMPLICATIONS

How to Find High-Quality Data Subsets?

The Central Challenge

Identifying or constructing **high-quality pre-training datasets** remains a central challenge in language model development. Modern approaches rely on various filtering strategies, but these are often empirical and heuristic.

Current Approaches

- **Language filters:** Restrict to target languages
- **Quality filters:** Remove low-quality content
- **fastText-based:** Language identification
- **Deduplication:** Filter near-duplicate content

Limitations

- Decisions largely **empirical and heuristic**
- Limited insight into what defines quality
- Often chosen based on **proxy benchmark performance**
- Dataset mixtures explored empirically

💡 This Work's Contribution

We take an **initial step toward elucidating the mechanisms** underlying pre-training data quality from an information-theoretic perspective and introduce a metric for evaluating the quality of pre-training subsets. Word-level unigram cross-entropy provides a **principled, computationally efficient** way to assess dataset quality without requiring expensive model training.

⚡

Efficient

No model training required

📊

Predictive

Correlates with performance

🔧

Practical

Easy to compute at scale

Future Direction: Develop more sophisticated metrics that capture beyond marginal word overlap, including contextual patterns, semantic similarity, and task-specific relevance.

IMPLICATIONS

Benchmarks Are Weakly Out-of-Distribution

Challenging Common Assumptions

A common assumption is that benchmark datasets **do not directly overlap with pre-training corpora** and are therefore highly out-of-distribution relative to the training data. Under this view, strong benchmark performance is primarily attributed to improved generalization.

The Traditional View

- ✕ **No Direct Overlap**
  Benchmarks unseen during training
- ✕ **Highly OOD**
  Large distribution shift
- ✕ **Generalization Required**
  Strong performance = good generalization

Our Findings

Implication for Benchmark Interpretation

Simply learning the statistical patterns present in the pre-training corpus can substantially contribute to improved benchmark performance. This calls for a reconsideration of how benchmark gains should be interpreted in the era of web-scale pre-training. Improved scores may reflect better alignment with training data rather than deeper understanding or generalization.

IMPLICATIONS

Cross-Entropy Is Not a Proxy for Benchmark Difficulty

Important Clarification

Word-level unigram cross-entropy measures **distributional affinity** between a benchmark and a pre-training corpus, **not the benchmark's intrinsic reasoning difficulty**. Cross-benchmark comparisons can be misleading for several reasons.

Benchmark Length

Benchmarks differ in length, making estimates **noisier and more sensitive to rare words** in shorter datasets.

Different Domains

Each benchmark has a **unique vocabulary distribution** that may align differently with pre-training data.

Task-Specific

Cross-entropy is **task-dependent** and should not be used as a universal difficulty metric.

⚠ Example: ARC Easy vs. Challenge

ARC Challenge is **widely considered harder** than ARC Easy, but this gap is **not reflected in their cross-entropy values:**

Challenge actually has slightly lower entropy!

Proper Interpretation

We interpret word-level unigram cross-entropy **primarily within a fixed benchmark** (e.g., comparing pre-training corpora for the same task), rather than as a universal scale across benchmarks.

Best Practice: Use cross-entropy to compare different pre-training datasets **on the same benchmark**, not to compare difficulty across different benchmarks.

Key Takeaway: Cross-entropy measures **alignment**, not difficulty. A benchmark can have low cross-entropy (good alignment) but still be challenging due to complex reasoning requirements, or vice versa.

CHAPTER 08

Conclusion

08

Key findings and future directions

CONCLUSION

Summary of Findings

Main Contributions

This work introduces **simple word-level overlap diagnostics** and evaluates how well they account for differences in zero-shot benchmark performance across pre-training corpora.

Finding 1: Robust Inverse Relationship

Across models, data scales, and tasks, we find a **robust inverse relationship** between word-level unigram cross-entropy and benchmark accuracy. Lower cross-entropy consistently predicts higher performance.

Finding 2: Weakly Out-of-Distribution

Many commonly used benchmarks are **only weakly out-of-distribution** relative to pre-training data. Simple word-overlap statistics can predict benchmark performance with surprising accuracy.

Finding 3: Word Frequency Matters

Word frequency statistics capture an essential aspect of data overlap that complements cross-entropy. Larger pre-training subsets with similar cross-entropy yield improved performance due to stronger learning signals.

Finding 4: Exceptions Exist

Some benchmarks (**BLiMP, MathQA, multilingual tasks**) do not follow this trend, requiring higher-order reasoning, abstract rules, or complex cross-lingual transfer beyond word overlap.

Core Insight

Benchmark performance often reflects statistical alignment between training and evaluation data rather than abstract generalization. This calls for a reconsideration of how benchmark gains should be interpreted in the era of web-scale pre-training.

FUTURE WORK

Key Takeaways & Future Work

Key Message

Word overlap between pre-training and benchmark datasets is strongly correlated with benchmark performance. This fundamental relationship has important implications for how we design, evaluate, and interpret language model benchmarks.

Open Challenges

- **Formalizing data quality:** What constitutes high-quality pre-training data beyond marginal word overlap?
- **Diagnostic benchmarks:** How to design evaluations that remain diagnostic under word overlap?
- **Generalization measurement:** How to truly measure abstract reasoning and understanding?

Future Directions

- **Sophisticated metrics:** Develop measures capturing contextual patterns and semantic similarity
- **Robust benchmarks:** Create evaluations resistant to word-overlap manipulation
- **True generalization:** Design benchmarks testing reasoning beyond statistical patterns

Implications for the Research Community

Benchmark Designers
Create more robust evaluations

Researchers
Interpret results with caution

Practitioners
Use cross-entropy for data selection

Final Thought: As language models continue to scale, understanding the relationship between training data and benchmark performance becomes increasingly critical. This work provides a foundation for developing more principled approaches to data selection and benchmark design.

THANK YOU

Questions & Discussion

Paper: arXiv:2502.17834

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Word Overlap | Benchmark Analysis | Data Quality